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BACKGROUND

Immunizations, across the globe, are heralded as one of the most successful public health interventions by averting numerous non-communicable and communicable diseases. However, in 2024, there were 14.3 million children missing out on any vaccination– so-called zero-dose children (WHO, 2025). In addition, targets for certain key vaccinations, such as the Hepatitis B vaccine, are missing the global Immunization Agenda 2030 targets (i.e., coverage of only 17% in the African Region). Across sub-Saharan Africa, the failure to expand vaccination services underscores a persistent crisis: access often remains tethered to socioeconomic status and residential stability, reinforcing deep-seated health disparities. This inequality contributes to multitudes of children at-risk for a wide spectrum of diseases that can cause severe health issues. Consequently, closing this immunization gap is contingent upon addressing the underlying economic and logistical determinants of health, without which the promise of global health equity will remain unfulfilled.

Supply (e.g., shortage of trained healthcare workers) and demand factors (e.g., maternal education) are often cited as the reason behind why there are low rates of immunization utilization. However, policy needs to provide more holistic approaches that are both cognizant of supply and demand factors while also incorporating individual and more complex underlying contextual factors as related to household wealth (Adebowale A. et al., 2019) to close equity gaps. By mapping these socioeconomic disparities and dismantling the structural barriers behind them, policymakers can design localized interventions that ensure healthcare delivery is dictated by actual need rather than systemic privilege.

RESEARCH QUESTION

What is the association between household wealth index and full childhood immunization coverage among children aged 12 to 35 months in selected sub-Saharan African countries?

- **Treatment/Exposure (X)**: Wealth Index. The Wealth Index measures household living standards based on consumer goods ownership and housing characteristics, derived using Principal Component Analysis (PCA). It is a composite indicator of living standards constructed via principal component analysis of household assets and housing characteristics.
- **Outcome (Y)**: Full Basic Immunization. A binary variable indicating whether a child received all basic antigens: 1 dose of BCG, 3 doses of OPV/IPV, 3 doses of DPT-HepB-Hib, and 1 dose of MR. Vaccination data were collected from written records or mother's recall.
- **Population**: Children aged 12-35 months in sub-Saharan Africa. Children aged 12-23 months should have completed basic immunization by 12 months; children aged 24-35 months should have additionally received a second dose of MR.

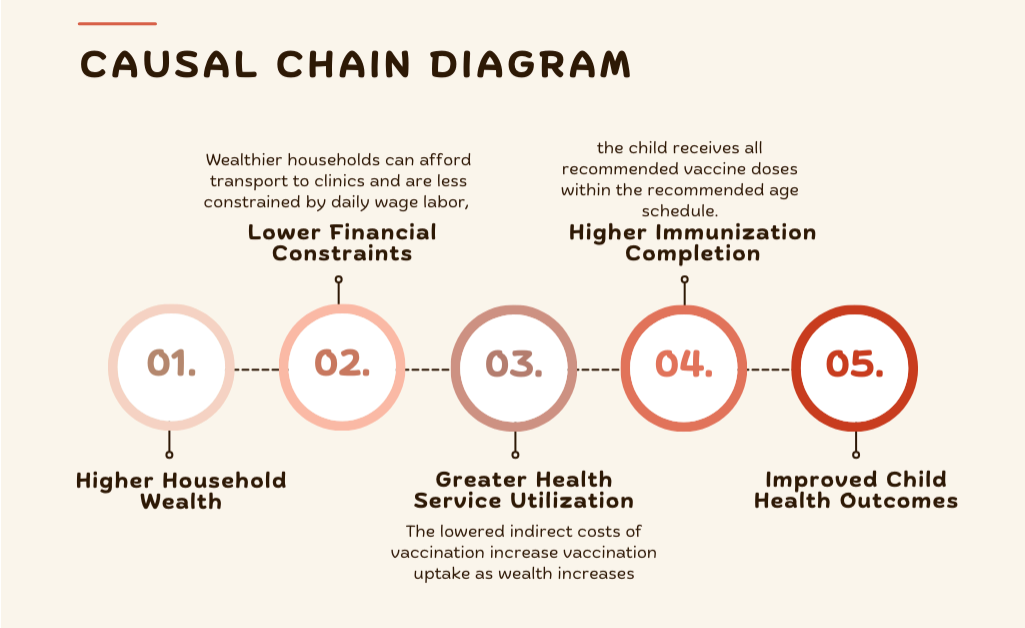
PROBLEM STATEMENT

Immunization is one of the most cost-effective public health interventions, yet coverage remains unequal across wealth groups in Sub-Saharan Africa. Children from poorer households are significantly less likely to complete recommended vaccination schedules. This disparity reflects several underlying market failures and behavioral barriers:

- **Credit Constraints**: Even when vaccines are provided free of charge, indirect costs (transportation, lost wages, waiting time) create financial barriers for poorer households.

- **Information Asymmetry**: Lower-income caregivers may have less access to accurate information about vaccine benefits and schedules.

THE CAUSAL CHAIN



KEY ASSUMPTIONS

Basic vaccine supply and facility functionality exist at minimum levels. Where severe supply-side failures persist (e.g., stockouts or non-operational clinics), the wealth gradient may be attenuated or disappear, as immunization becomes supply-constrained rather than demand-constrained Wealth affects immunization primarily through access and information rather than strong cultural resistance. Confounding factors (e.g., maternal education, urban residence, regional infrastructure) are appropriately controlled for in analysis.

LITERATURE REVIEW

The immunization gap across sub-Saharan Africa has made understanding the role of household wealth in childhood vaccination increasingly important. Existing research highlights two primary perspectives on this relationship. The first and more dominant view is that economic inequality serves as a structural barrier that limits access to and affordability of vaccination services, thereby contributing to persistent coverage gaps between low- and high-income households. The second perspective suggests that in some contexts, immunization programs have been implemented in ways that more effectively target poorer households, resulting in equal or even higher vaccination coverage among children from low-income families.

In their landmark cross-sectional study of 153,632 children across sub-Saharan Africa, Mutua and colleagues (2021) provide nearly two decades of evidence that lower household wealth is significantly associated with reduced on-time vaccination coverage among children aged 12 to 36 months. Similarly, evidence from a study of nine sub-Saharan African countries shows a 11.7 percentage point gap in full immunization coverage between children aged 12 to 23 months in rich and poor households (Fenta et al.,

2021). However, this sample is somewhat dated, drawing on data from 2013 to 2017, and is limited to nine sub-Saharan African countries, which may make it less relevant to examining current public health conditions and regional generalizability. Using more recent data from 2016 to 2023, Jejaw and colleagues (2025) find that children aged 12 to 23 months whose mothers live in low community-level wealth areas have the highest prevalence of missed vaccination opportunities and are 9% more likely to experience missed vaccinations compared to their counterparts in wealthier communities. Yet, this study includes only 20 sub-Saharan African countries, representing less than half of the region's 49 countries, which may limit the generalizability of its findings.

One study posits that household wealth is associated with higher overall childhood vaccination rates in certain countries. Research by Bobo et al. (2021) finds that although most sub-Saharan African countries exhibit substantial pro-rich inequality in immunization coverage, The Gambia and Namibia demonstrate pro-poor coverage, possibly due to more effective targeting of vulnerable populations in smaller countries. However, the researchers admit that the mechanisms behind this pattern remain unclear. Overall, the existing literature suggests a generally positive association between household wealth and childhood immunization coverage, with some caveats. Findings differ across countries and study populations, with some analyses documenting modest wealth-based gaps and others identifying reversed or attenuated disparities in specific settings. However, much of the current literature relies on pre-pandemic data, focuses on children aged 12 to 23 months, and examines only a subset of sub-Saharan African countries, limiting regional representativeness and contemporary relevance (such as the USAID shutdown). This study addresses these gaps by evaluating the relationship using more recent data and a broader set of sub-Saharan African countries to provide a more regionally representative assessment.

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